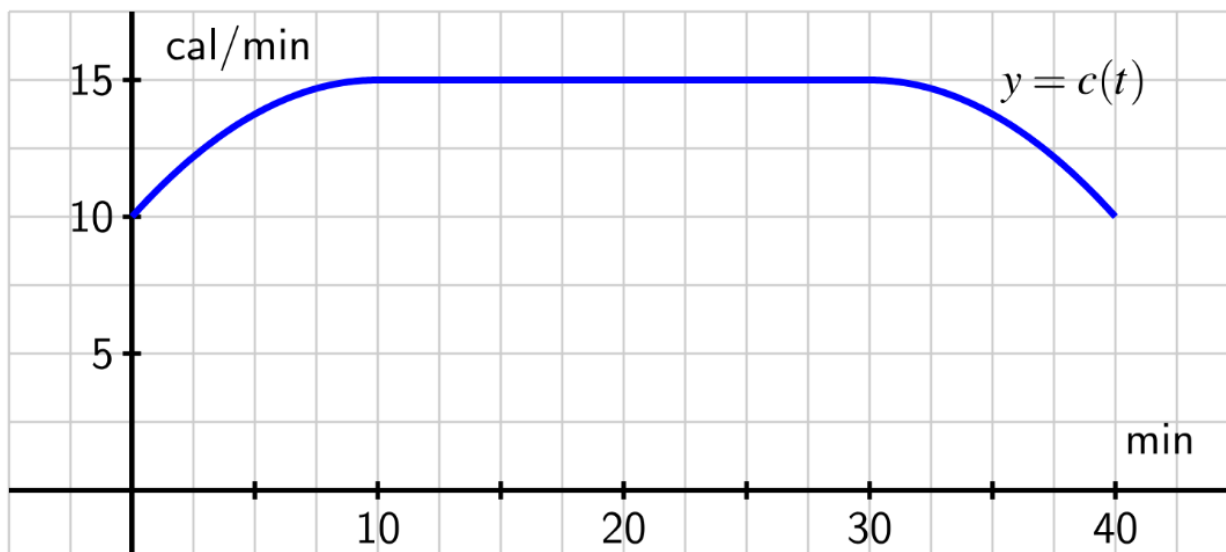


**PS12: Definite integrals and the Fundamental Theorem of Calculus**

During a 40-minute workout, a person riding an exercise machine burns calories at a rate of  $c(t)$  calories per minute, where the function  $y = c(t)$  is given by the following information:

- On the interval  $0 \leq t \leq 10$ , the formula is  $c(t) = -0.05t^2 + t + 10$  (warmup);
- on the interval  $10 \leq t \leq 30$ , the formula is  $c(t) = 15$  (conditioning phase);
- on the interval  $30 \leq t \leq 40$ , the formula is  $c(t) = -0.05t^2 + 3t - 30$  (cooldown).

Here's a graph of  $c(t)$ .



1. (IN1) Shade in the area under  $c(t)$  between  $t = 10$  and  $t = 30$ . Use some simple geometry to calculate this area. Give units.
2. (INx) Write a sentence explaining what the answer to part 1 *means* in the context of the person exercising.
3. (IN2) Use a Riemann sum with 4 rectangles to approximate the area under  $c(t)$  between  $t = 30$  and  $t = 40$ :
  - How wide should each rectangle be?
  - Decide on a consistent way to choose the height of each rectangle.  
(Do you want to do the top-left corner? the top-right corner? the middle? Up to you!)
  - Sketch your four rectangles on the graph of  $c(t)$ .
  - Compute the four rectangle areas separately, and give units.
  - Add up the four rectangle areas to get a total area.
  - Based on your height choices, is your estimate an *overestimate* or an *underestimate* of the actual area? Why?

4. (IN3) Find a formula for an antiderivative  $C(t)$  for the portion of  $c(t)$  that's on the interval  $0 \leq t \leq 10$ .
5. (IN5) Use your antiderivative  $C(t)$  to find the exact value of  $\int_{t=0}^{t=10} c(t) dt$ . Give units.
6. (IN5) Now find the *exact* value of  $\int_{t=30}^{t=40} c(t) dt$ .
  - Careful: you'll need to find a new antiderivative, because the formula for  $c(t)$  is different!
7. Put it all together: Find the *exact* value of  $\int_{t=0}^{t=40} c(t) dt$ , give units, and explain what this number means in the context of the person exercising.